

The Digital Transformation Competences for Brazilian Automotive Managers: A Transdisciplinary Engineering Approach

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Abstract. New technologies related to Digital Transformation (DT) and the Industry 4.0 (I4.0) modify the way business and productive processes are carried out, generating complex changes for industry and engineering, establishing new tasks and human roles, and interacting with the characteristics of Transdisciplinary. Digital engineering managers play an integrative role by relating and using the organisation's digital technological knowledge to generate better business results. The characterization of managers' competences to guide and stimulate value creation in industrial sectors is still not sufficiently investigated and emerges as a critical element for industrial development in the digital age. This research fulfils this gap and aims to rank four types of necessary competences for engineering managers facing the DT/I4.0 in the automotive sector. The methodological approach adopted is quantitative, based on the judgement of engineering managers from the Brazilian automotive sector, which is globally representative in terms of productivity. An Analytic Hierarchy Process (AHP) is applied in the data treatment. Results are based on a sample of 35 interviews from six automotive companies with different levels of complexity in production operations and formal programs for DT/I4.0 implementation. Findings indicate the relative priority for the digital technical, managerial, social, and motivational competences, presenting insights with implications to guide the development of the digital engineering managers.

Keywords. Transdisciplinary engineering, Digital transformation, Engineering managers, Competence, Automotive Sector.

Introduction

The Digital Transformation (DT) and the Industry 4.0 (I4.0) emerge as a challenge for companies in dynamic and highly competitive markets [1]. New technologies modify the way business and productive processes are carried out, establishing new tasks and human roles both in characteristics and content [2][3]. I4.0 generates complex changes for industry and engineering, which can influence or relate to the characteristics of Transdisciplinary [4], and affect the needed competences for engineering managers [2].

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Digital managers play an integrative role by articulating and applying the organisation's digital technology knowledge to generate better results [5][6]. Their action emerges as a critical element for industrial development in the digital age [7][8]. However, the required managerial competences to guide value creation in industrial sectors is still not sufficiently investigated. The DT/I4.0 impact managerial tasks, indicating aspects to be further investigated in different industrial contexts, including:

- the identification and characterization of managers' competences [2][5];
- the understanding of the perspectives and expectations of industrial managers, regarding changes in role and competences [9][7];
- the priority for developing managers' competences [10];
- identification of which managerial competences are capable of providing appropriate changes for DT/I4.0 in different industrial contexts [5][6];
- the understanding of managers role as team leaders in a DT/I4.0 context [9].

In this research, the main objective is to rank four types of necessary competences for engineering managers facing the DT and I4.0 implementation in the automotive sector. A quantitative approach is adopted, based on the judgement of engineering managers from the Brazilian automotive sector, and the Analytic Hierarchy Process (AHP) is applied in the data treatment. The automotive industry has been a pioneer in introducing new production technologies and work organisation practices, which have been replicated in other sectors [11], and the Brazilian automotive industry is the 8th largest in the world and produced 2,369,769 vehicles in 2022 [12]. Thus, it provides a fertile environment for research, as their managers adapt to technological and business challenges, in a cultural and institutional context that influences the technological transformation. The results are based on a sample of 35 interviews from six automotive companies with different levels of operational complexity and formal programs for DT/I4.0, implementation. The relative priority for digital technical, managerial, social, and motivational competences are analysed. The managers' perception, both at tactical and strategic level, can generate results, which, aligned with the technological context of the sector and business characteristics, can contribute for more appropriate management competence models and provide insights to guide the development of managers.

This work contributes to the understanding of changes in the nature of engineering in the face of DT and I4.0, analysing competences that can guide the development of modern engineering managers to play an integrative role in combining people, disciplines, methods and implemented technological knowledge in a transdisciplinary path to solve current complex problems and generate better results.

This paper is structured presenting the related theoretical background in Section 1. Section 2 presents the adopted methodology. The results and discussion are detailed in Section 3, supporting the conclusion finally presented in Section 4.

1. Theoretical background

1.1. Digital Transformation and Industry 4.0

The implementation of new technologies, such as cyber-physical systems, big data, additive manufacturing, internet of things, artificial intelligence, robotics and cloud computing [13], indicate a transformation in the production and distribution of goods and

services. It generates possible far-reaching consequences, on issues ranging from the development of workers/managers competences to environmental impacts, income distribution and social well-being [14].

The DT/I4.0 emerged as a new industrial revolution, considering the capacity of new technologies to create intelligent and autonomous systems capable of producing customised products in small batches at low costs [15], improve working conditions by automating repetitive tasks, and ultimately bring rapid interaction with customers and suppliers and organisational performance [16]. It implies a manufacturing system in which digitally enabled machines execute routines while interacting with operators and other machines through the internet of things [17]. Although it can be considered as an application of digital technology in manufacturing, its adoption requires human-machine interaction [18]. Therefore, DT/I4.0 should not be seen only as a technological issue increasing productivity, as it also impacts the organisation of work [19].

The implementation of DT/I4.0 then demands a qualified workforce, changes routines and affects planning and decision-making to produce personalised, high-quality products and services [20]. Its implementation can generate impacts at the individual (professionals) or organisational level, in systems and work teams, in management practices, and in innovation and production networks [21]. It demands new competences from professionals, which can be classified in relation to the type of task (related to the level of decision, domain of knowledge, interdisciplinarity, etc.) to behavioural and cognitive aspects (related to responsibility, systematic thinking, etc.) or social (in terms of flexibility and adaptability) [22].

The adoption of DT/I4.0 technologies can positively influence work flexibility and autonomy, also leading to better production performance, increased creativity and innovation [22]. However, the acquisition and dissemination of technological knowledge within the organisation is important in supporting the process of technological implementation and innovation [23][24]. In addition to the human interaction with technology itself, the implementation of DT/I4.0 requires the relationship with operational security resources and protocols, learning, training, and continuous performance analysis [25]. Although DT/I4.0 technologies affect individual and group learning, reinforcing the need for updated training strategies, related literature still places greater focus on technological issues [26], generating a demand for initiatives that explore more human aspects, such as the competences of managers for I4.0 and management practices related to the industrial context [17].

1.2. Managers competences in the DT/ I4.0

The establishment of a relationship between technologies and human competences constitutes a challenge for industrial ecosystems in terms of developing and converting innovative potential to deal with opportunities and challenges of DT/I4.0 [5].

DT/I4.0 demand new mentalities and competences depending on each specific organisational level, and at the leadership level it requires greater ability to adapt to constant changes, and must also be able to stimulate teams of workers to this condition [27]. The changes generated by the technological transformation impact leaders and managers at the tactical and strategic levels of organisations, who need to develop the appropriate related competences to be able to implement strategies aligned to the new context [28]. The digital training of managers constitutes a critical aspect for DT/I4.0 and requires the development of competences linked to the technological opportunities.

At the same time, managers are also challenged to constantly develop new social and motivational competences, to promote engagement and creativity on work teams [1].

The development of managers' competences requires a change of mindset regarding motivation to adopt technologies, continuous learning on digital technologies, commitment to keeping updated digital information, innovative thinking and attitudes and the development of a vision aligned with technological opportunities [29]. Managers at a strategic tactical level will act as moderators of technological power in the management of related resources, guiding and controlling processes, new relationships and operations strategies, and exerting behavioural influence on people involved in new tasks [6]. Therefore, managers must guide people's interaction with technological resources efficiently, through the dynamic and structural management of digital production technologies, which also requires a certain level of knowledge and competence about new technologies, to support the company's initiatives and strategies to obtain good performance [5].

Digital technologies create opportunities for new configurations in the industrial environment and for the collaborative interactions of digital-human activities [30][37]. It requires that managers assume the role of directing efforts, developing and integrating such configurations for productivity or innovation processes [31][38].

The concept of competence is related to the behaviour of individuals in carrying out activities efficiently, aiming to achieve satisfactory performance, generally involving knowledge, skills and attitudes necessary to carry out routine activities [27]. The competences of managers in the digital technological context guide the development of diversified product and service projects digitally aligned with customers, technology strategies, marketing, communication and operations [32].

Gagné-Pratte, Mosconi and Jorge [33], propose four types of managers' digital competences: 1) 'digital technical competences', related to work skills, knowledge and attitudes, which individuals develop through practice or learning digital technologies; 2) 'digital managerial competences', related to the skills, knowledge and attitudes necessary for managers to make informed decisions and solve problems effectively, supported by digital technologies; 3) 'digital social competences', related to skills, knowledge and attitudes linked to the social values of individuals, in collaboration and supported by digital technologies; and, 4) 'digital motivational competences', related to the skills, knowledge and attitudes to motivate the workforce and ensure people's active involvement in the success of technological transformation.

2. Method

The AHP [34] is applied to rank four types of competences adapted from Gagné-Pratte, Mosconi and Jorge [33], considering the perceptions of engineering managers facing the DT/I4.0. First based on the digital literacy competences model [33], the four types of competences integrating this investigation were identified as relevant, considering the updated literature stated about industrial and engineering managers.

The AHP method is chosen due to its capacity to combine the interviewees perceptions regarding different kind of factors, considering individual and group levels of importance, also providing a base for future applications of other quantitative or qualitative methods, such as semi-structured interviews. The decision-making problem was then converted into a questionnaire of pairwise comparisons. The relative priority for digital technical, managerial, social, and motivational competences were analysed.

2.1. Data collection and analysis

AHP [34], is a multi-criteria decision method that decomposes decision-making problems (i.e. with multiple options) into a hierarchical structure, considering the main purpose on the top (Level I). Figure 1 presents the decision-making structure, considering the four types of competences (Level II), considering digital technical, managerial, social and motivational competences, as adapted from Gagné-Pratte, Mosconi and Jorge [33].

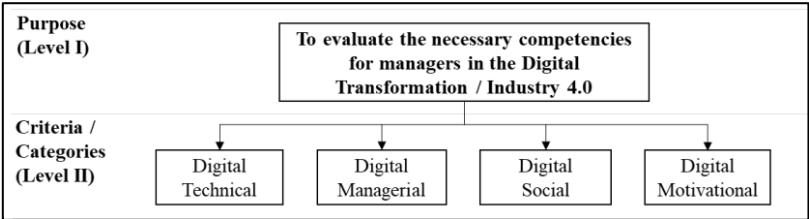


Figure 1. Hierarchical structure of competences - Adapted from Gagné-Pratte, Mosconi and Jorge [33].

The decision-making structure (Figure 1) was converted in a questionnaire of pairwise comparisons, used to capture the perception of the engineering managers about the relative importance of the competences, taking the Saaty's fundamental scale [34] as a parameter. The Incomplete Pairwise Comparison (IPC) [35] was used to reduce the minimum required number of comparisons and the final questionnaire was composed of three pairwise comparisons. Figure 2 presents a pairwise comparison example.

In order to better guide your organization to thrive in the digital transformation / Industry 4.0 context, based on the established scale, compare and indicate the relative importance of the following alternatives:
Tip: hover alternatives text to see the detailed definition.

	9	7	5	3	1	3	5	7	9	
Digital <u>Technical</u> Competencies	☐	☐	☐	☐	☒	☐	☐	☐	☐	Digital <u>Managerial</u> Competencies

Figure 2. Questionnaire pairwise comparison example.

Respondents were personally contacted and introduced to the research purpose, questionnaire and content related to the evaluated competences. They were informed about the ethical aspects of data collection and answered the questionnaire spontaneously. The questionnaires were answered in an online digital platform for data collection. The average response time was 8 minutes. During the response process, respondents were provided by the access to the definition of each type of competence analysed.

After data collection, the relative importances attributed by each respondent were entered into a reciprocal value matrix and the priorities of the factors were calculated by the AHP method [34]. Result consistency verification was performed by AHP and IPC [35], and presented high consistency indexes ($CI = 0$).

2.2. Sample characterization

Results are based on a sample of 35 interviews from six automotive companies, three Automakers and three Tier 1 suppliers, with different operations complexity levels. The selected companies are well established and have formal programs for DT/4.0 implementation. Table 1 presents some of their characteristics.

Table 1. Companies' characterization.

Automotive Company	Level	Main products	Number of respondents
AC1	<i>Tier 1</i>	Metal Components / Assemblies	1
AC2	Automaker	Passenger Vehicles	7
AC3	<i>Tier 1</i>	Structural Components	8
AC4	<i>Tier 1</i>	Engines	10
AC5	Automaker	Passenger Vehicles	3
AC6	Automaker	Trucks / Buses	6

Respondents were selected by convenience, considering their routine relevance in terms of DT/I4.0 implementation and adaptation. The sample is composed of highly experienced engineering managers, in different hierarchical levels, in charge of teams, industrial engineering areas or processes undergoing technological transformation. Table 2 summarises the characteristics of the selected respondents.

Table 2. Respondents' characterization.

Job function	Number of Resp.	Age			Experience			
		Up to 30 years	31-40 years	More than 40 years	Up to 10 years	11-20 years	21-30 years	More than 30 years
Director	3	0%	0%	100%	33%	67%	0%	0%
Manager	19	26%	42%	32%	74%	26%	0%	0%
Division Manager	2	0%	0%	100%	0%	50%	50%	0%
Project Manager	4	0%	75%	25%	100%	0%	0%	0%
Leader	7	0%	43%	57%	57%	29%	0%	14%
<i>Overall</i>	35	14%	40%	46%	66%	28%	3%	3%

3. Results and discussion

The 'digital managerial competence' is indicated as the main priority (36.0%) for the development of engineering managers (Table 3). This competence is in general related to the capacity of managers to deal with problem solving or decision-making aspects, using technologies as supportive resources to support related practices and generate better performance. Routine problems in the new technological context requires engineering managers to analyse a situation, identifying, evaluating, selecting and using digital technologies and possible technological responses to solve a given task or problem, by having an inquisitive mindset, curious and passionate attitude towards analytics-based business problems [33]. DT/I4.0 implies a dynamic routine of continuous change, which requires industrial and engineering managers to have a greater ability to constantly adapt to deal with new and different problems [27][38]. This dynamism can stimulate managers to find new ways, also supported by the new technologies, to innovate and find new appropriate solutions [23]. The transdisciplinary engineering concepts [4] can establish a link with this new technological context and help engineering managers to find new ways for problem solving.

Table 3. Summary of competences priorities.

Criteria and Alternatives (Competences)	Priorities by Automotive Company ² (%)						Overall ³ (%)
	AC1	AC2	AC3	AC4	AC5	AC6	
Digital Technical	6.1	34.1	20.4	19.6	23.4	20.7	20.7
Digital Managerial	42.7	37.1	21.1	44.6	40.1	30.3	36.0
Digital Social	8.5	12.8	23.2	16.7	8.0	9.3	13.1
Digital Motivational	42.7	16.1	35.4	19.0	28.5	39.7	30.2
Respondents #	1	7	8	10	3	6	35

The decision-making process in the DT/I4.0 context also requires the use of the new technologies to make choices based on data, information, knowledge and experience, assessing alternative solutions, using critical thinking, understanding the connections between ideas, identifying the relevance of arguments, detecting errors in reasoning, and trusting your intuition and facts, to guide strategic business decisions that align with the organisation's goals and initiatives [33]. The literature indicates that the implementation of DT/I4.0 dynamically affects the routine industrial planning, to produce personalised, high-quality products and services demanding from managers a decision-making process also aligned to the new technological context [20]. New technologies, such as big data and artificial intelligence are expected to generate valuable information to support managers and facilitate the decision-making process [13]. As a direct implication, the development of 'digital managerial competences' by engineering managers, aligned with the new technological context, is expected to promote a better combination of people, disciplines, methods and implemented technological knowledge in a transdisciplinary path to solve current complex problems and generate better results.

The 'digital motivational competence' was also indicated in a high priority level (30.2%), indicating that even considering the advent of new technologies, human capabilities are important to deal with the new industrial context and have to be considered for the engineering managers competences development. In the new technological context, the 'digital motivational competence' is related to the leadership capacity, supported by new technological opportunities, to motivate the workforce and ensure people's engagement [33]. DT/I4.0 challenged managers to constantly develop the motivational competences, to generate and stimulate the engagement and creativity of work teams [1]. Aligned with the technological transformation dynamism, managers are demanded to motivate the workforce and integrate digital resources in the work environment, using digital platforms to communicate changes, predicting or perceiving quickly evolving business needs, being adaptive in the challenging technological context while learning quickly, managing/mediating changes and innovation, risks and unexpected crises, and tools to rapidly shift strategies [33]. DT/I4.0 requires the leadership level greater ability to adapt to constant changes, and they must also be able to stimulate teams of workers to this condition [27], to attain better production performance, increased creativity and innovation [22]. This leadership characteristic requires from managers a human capacity to inspire, direct and mobilise individuals to use vision-setting definition, strategy development and goal achievement, to develop an

² According to the 'Aggregating individual judgments' (AIJ) method [36], which consolidates the data aggregating individual judgement, through calculation of the geometric mean, generating a group judgement. The results consistency is based on the AHP-Incomplete Pairwise [35].

³ According to the 'Aggregating Individual Priorities' (AIP) method [36], which consolidates data from the automotive companies, considering the aggregation of the obtained priorities for each company, through calculation of the arithmetic average, generating a synthesis of the general results.

orientation and guide the organisation towards DT/I4.0, for the digital technology adoption in sustainable ways [33][37]. The literature indicates that leaders at the tactical and strategic levels of organisations, such as engineering managers, need to be able to implement strategies aligned to the new context [28], so ‘digital motivational competences’ can imply in efficient ways to guide people for the technological adaptation, exploring the major potential of individuals and groups.

Even identified in lower levels, ‘digital technical’ and ‘digital social’ competences remains relevant for the engineering managers development, contributing to constitute more adequate professional profiles to face the challenges of DT/I4.0.

4. Conclusion

This research aimed to rank four types of competences for engineering managers facing the DT/I4.0 in the Brazilian automotive sector, globally representative in productivity. A quantitative approach (AHP) was adopted, based on the engineering managers judgement. A sample of 35 interviews from six automotive companies with different levels of complexity in production operations, with formal programs for DT/I4.0 was analysed. The relative priority between technical, managerial, social, and motivational competences were identified.

The ‘digital managerial competence’, generally related to the capacity of managers to deal with problem solving or decision-making aspects, using technologies as supportive resources to support related practices and generate better performance [33], was identified as the main priority for the development of engineering managers facing the technological transformation in the analysed automotive companies. The dynamic routine of continuous change implied by DT/I4.0 reinforce the importance of the managerial competence and requires engineering managers to have a greater ability constantly adaptation to deal new and different problems [27], to find new ways, supported by the new technologies, to innovate and find new appropriate solutions [23], and develop a decision-making process aligned to the new technological context [20]. The transdisciplinary engineering view is expected to establish a link with this new technological context and help engineering managers to find new managerial ways for problem solving and decision making.

The ‘digital motivational competence’ was also identified in a high priority level, indicating that even considering the advent of new technologies, human capabilities are important to deal with the new industrial context and have to be considered for the engineering managers competences development. DT/I4.0 bring challenges for managers that need to constantly develop the motivational competences, to generate and stimulate the engagement and creativity of work teams [1].

The perception of the analysed industrial engineering managers, integrating tactical and strategic levels, aligned with the technological transformation context of the Brazilian automotive sector, contribute to establish a more appropriate management competence model, presenting insights with implications to guide the development of these managers. The proposed AHP model can be applied by human resource practitioners in other industrial sectors or contexts as an instrument to diagnose the needs for managers development during the technological transition.

As future research opportunities, qualitative analysis about the analysed digital competences, as well as about the identified priorities, could help to clarify the related reasons and how these competences interact supporting engineering managers in the new

technological context. Skills, knowledge or attitudes related to the four analysed competences can also be considered and inserted in sublevels of the decision-making structure. The methodological approach can also be applied in other automotive samples, industrial sectors or contexts to identify the related competence' priorities.

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